YEAR 11 PHYSICS, AQA GCSE, PART 1 OF 4

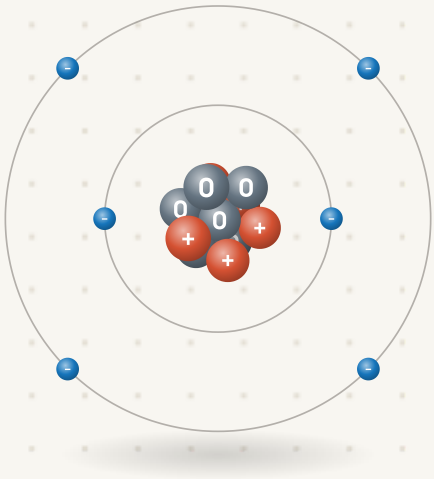
Atoms & Nuclear Radiation.

Part 1 of 4 Lessons 1–2

IN THIS BOOKLET

- Lesson 1: Inside the atom
- Lesson 2: Isotopes and ions

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

How this booklet works.

This unit starts from the atoms you met in Key Stage 3 and in chemistry, then asks how we know what is inside them and what happens when a nucleus is unstable. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations and diagrams. Fill in the slide examples with your teacher, then try "Your turn" alone.

TRY IT
Practice that builds up, then a challenge. Quick checks from the slides follow some lessons.

LAB
Record what you find in a simulation or a model experiment.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

The route through the unit (this booklet highlighted)

01	**Inside the atom** Slides 3–6	02	**Isotopes and ions** Slides 7–8
03	From plum pudding to the nucleus	04	Energy levels and the nucleus
05	Radioactive decay	06	Nuclear equations
07	Properties and uses of radiation	08	Half-life and random decay
09	Half-life calculations	10	Contamination and irradiation

Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

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LESSON

01 Inside the atom.

AQA Physics 4.4.1.1–4.4.1.2; Combined Science 6.4.1.1–6.4.1.2

BIG QUESTION

Everything around you is made of atoms, yet an atom is almost entirely empty space. So what is actually in there?

BY THE END YOU CAN

- describe the charge, mass and position of protons, neutrons and electrons
- state the size of an atom and of its nucleus
- explain why an atom has no overall charge
- use atomic and mass numbers to find the numbers of each particle

atom · nucleus · proton · neutron · electron · atomic number · mass number

SLIDES AND SIMULATIONS

Teaching deck, slides 3–6

Three particles, the size of an atom, and reading the symbol.

panphy.app/gcsephy/decks/atoms-and-radiation/#3

RETRIEVE

What do you already know?

From Key Stage 3 and chemistry. Have a go.

1 What is everything around you made of?

Atoms (tiny particles).

2 Name the three particles found in atoms.

Protons, neutrons and electrons.

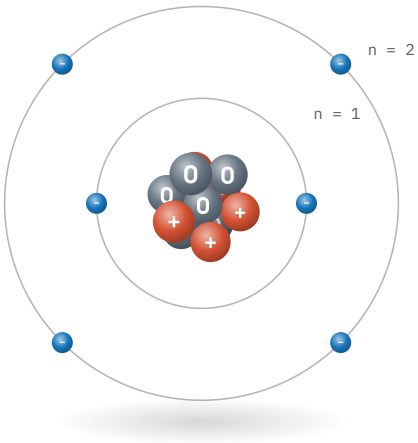
3 What is an element?

A substance made of only one type of atom.

EXPLAIN

What is inside an atom?

Three particles



A carbon atom: 6 protons and 6 neutrons in the nucleus, 6 electrons around it.

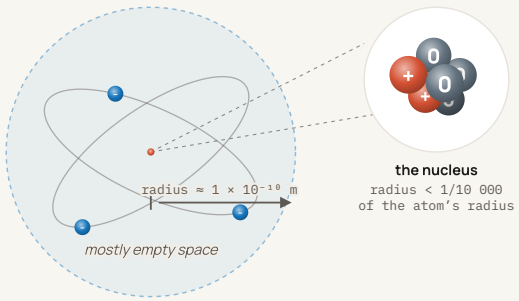
At the centre of every atom is a tiny **nucleus**. It contains **protons**, which are positive, and **neutrons**, which have no charge. So the nucleus is positive.

Negative **electrons** are arranged at different distances from the nucleus, called **energy levels**.

Particle	Relative charge	Relative mass	Where it is
Proton	+1	1	in the nucleus
Neutron	0	1	in the nucleus
Electron	−1	very small	in energy levels around the nucleus

! An atom has **no overall charge**: it has the **same number of electrons as protons**, so the positive and negative charges cancel. Neutrons have no charge, so they do not count.

Mostly empty space



Not to scale: a real nucleus is far too small to draw.

An atom has a radius of about **1×10^{-10} m** (0.000 000 000 1 m).

The radius of the nucleus is **less than 1/10 000** of the radius of the atom. Yet **almost all the mass** of the atom is in the nucleus, because protons and neutrons are far heavier than electrons.

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THINK OF A FOOTBALL STADIUM

If an atom were as big as a football stadium, the nucleus would be about the size of a pea on the centre spot. The electrons would be somewhere in the stands. Everything in between is empty space.

Where the model breaks: a pea is light, but the nucleus holds almost all the mass; and electrons are not little balls sitting in seats.

WORKED EXAMPLE

An atom has a radius of 1×10^{-10} m. Its nucleus is 1/10 000 of this. Calculate the radius of the nucleus.

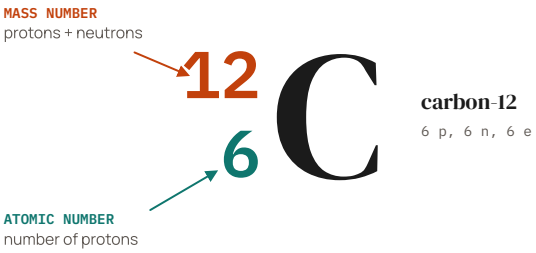
KNOW	atom radius = 1×10^{-10} m
METHOD	divide by 10 000 = 1×10^{4}
WORKING	$1 \times 10^{-10} \div 1 \times 10^{4}$
ANSWER	**1×10^{-14} m**

YOUR TURN

An atom has a radius of 3×10^{-10} m. Its nucleus is 1/10 000 of this. Calculate the radius of the nucleus.

KNOW	atom radius = 3×10^{-10} m
METHOD	divide by 1×10^{4}
WORKING	$3 \times 10^{-10} \div 1 \times 10^{4}$
ANSWER	3×10^{-14} m

Reading the symbol



- The **atomic number** (bottom) is the number of **protons**. It decides which element the atom is.
- The **mass number** (top) is the number of **protons + neutrons**.
- Number of neutrons = mass number − atomic number.
- In an atom, number of electrons = number of protons.

WITH YOUR TEACHER SLIDE 6

How many protons, neutrons and electrons are in an atom of $^{23}_{11}$Na?

PROTONS	atomic number = 11
NEUTRONS	23 − 11 = 12
ELECTRONS	same as protons = 11
ANSWER	11 p, 12 n, 11 e

YOUR TURN

How many protons, neutrons and electrons are in an atom of $^{27}_{13}$Al?

PROTONS	atomic number = 13
NEUTRONS	27 − 13 = 14
ELECTRONS	same as protons = 13
ANSWER	13 p, 14 n, 13 e

PRACTISE

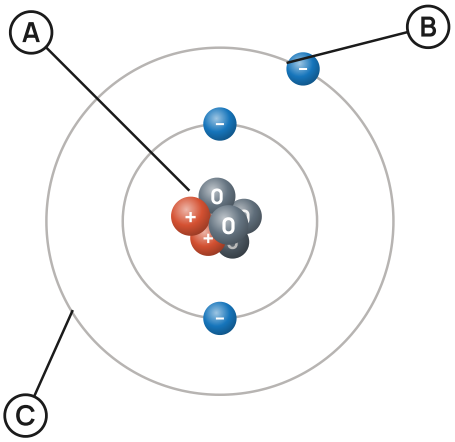
Try it

1 Complete the sentences using the word bank.

At the centre of an atom is a tiny **nucleus**. It contains **protons** and **neutrons**. **Electrons** are arranged in energy levels around it. A proton has a relative charge of **+1** and an electron has a relative charge of **−1**. Almost all of the **mass** of the atom is in the nucleus.

WORD BANK: nucleus · protons · neutrons · Electrons · mass · +1 · −1 · 0 · charge

2 The diagram shows a lithium atom. Name the parts labelled A, B and C. Then say which particles are inside part A.



A **nucleus**

B **electron**

C **energy level**

Inside A: **3 protons, 4 neutrons**

3 SUPPORTED Complete the table.

Symbol	Protons	Neutrons	Electrons
$^{4}_{2}\mathrm{He}$	2	2	2
$^{7}_{3}\mathrm{Li}$	3	4	3
$^{16}_{8}\mathrm{O}$	8	8	8
$^{56}_{26}\mathrm{Fe}$	26	30	26
$^{238}_{92}\mathrm{U}$	92	146	92

4 ON YOUR OWN An atom has 17 protons and 18 neutrons. Write its symbol in the form $^{A}_{Z}\mathrm{X}$. The element with 17 protons is chlorine, Cl.

$^{35}_{17}\mathrm{Cl}$ (mass number 17 + 18 = 35)

5 Which part of an atom contains almost all of its mass? Tick **one** box.

- ☐ the electrons
- ☑ the nucleus
- ☐ the energy levels
- ☐ the empty space

6 Explain what is wrong with Leo's statement, and correct it.

> **Leo:** An atom is neutral because it has the same number of protons and neutrons.

Neutrons have no charge, so they do not balance anything. An atom is neutral because it has the same number of electrons (−) as protons (+), so the charges cancel.

7 CHALLENGE The radius of an atom is about 10 000 times the radius of its nucleus. If you drew the nucleus as a dot 1 mm across, how wide would you have to draw the atom? Give your answer in metres, and say whether it would fit on this page.

10 000 × 1 mm = 10 000 mm = 10 m. It would not fit on the page: this is why atom diagrams are never to scale.

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: A 'describe' question about structure needs where each particle is, not just its name.

8 Describe the structure of an atom. **[4 marks]**

A tiny nucleus at the centre (1), containing protons and neutrons (1). Electrons are at different distances from the nucleus / in energy levels around it (1). Most of the atom is empty space: the nucleus radius is less than 1/10 000 of the atom's (1).

9 An atom of potassium is $^{39}_{19}\mathrm{K}$. (a) Give the number of neutrons. (b) Explain why the atom has no overall charge. **[3 marks]**

(a) 39 − 19 = 20 (1). (b) It has 19 protons and 19 electrons (1); their charges are equal and opposite, so they cancel (1).

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- The nucleus contains protons (+1) and neutrons (0). Electrons (−1) are in energy levels around it.
- Atom radius ≈ 1×10^{-10} m; nucleus radius less than 1/10 000 of this.
- Almost all the mass is in the nucleus.
- Atomic number = protons. Mass number = protons + neutrons.
- In an atom, electrons = protons, so there is no overall charge.

COVER THE PAGE AND ANSWER

1 What is the relative charge of a neutron?

2 What is the radius of an atom, roughly?

3 What does the mass number tell you?

4 How many neutrons are in $^{31}_{15}\mathbf{P}$?

I CAN…

- describe the charge, mass and position of each particle
- state the size of an atom and of its nucleus
- explain why an atom is neutral
- work out protons, neutrons and electrons from nuclear notation

LESSON

02 Isotopes and ions.

AQA Physics 4.4.1.2; Combined Science 6.4.1.2

BIG QUESTION

Carbon-12 and carbon-14 are both carbon, but only carbon-14 is radioactive. What is different about them?

BY THE END YOU CAN

- explain what isotopes are
- compare isotopes using nuclear notation
- describe how an atom becomes a positive ion
- say what changes, and what does not, when an ion forms

isotope | ion | positive ion | outer electron | stable | unstable

SLIDES AND SIMULATIONS

Teaching deck, slides 7–8

Isotopes of carbon; a lithium atom becoming an ion.

panphy.app/gcsephy/decks/atoms-and-radiation/#7

RETRIEVE

Do now

From earlier lessons. No notes!

1 What is the relative charge of a neutron?

0

2 How many neutrons are in $^{31}_{15}P$?

31 − 15 = 16

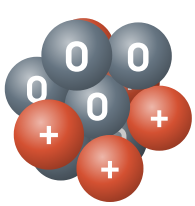
3 Why does an atom have no overall charge?

It has the same number of electrons as protons.

EXPLAIN

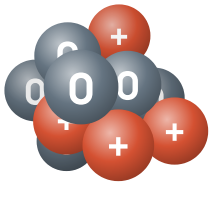
Same element, different atoms

Isotopes



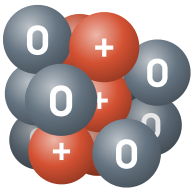
carbon-12

6 p, 6 n



carbon-13

6 p, 7 n



carbon-14

6 p, 8 n

Isotopes are atoms of the **same element** with **different numbers of neutrons**. They have the same number of protons, so the same atomic number, but different mass numbers. We name an isotope by its mass number: carbon-12, carbon-13, carbon-14.

Isotope	Symbol	Protons	Neutrons	Stable?
carbon-12	$^{12}_{6}C$	6	6	stable
carbon-13	$^{13}_{6}C$	6	7	stable
carbon-14	$^{14}_{6}C$	6	8	unstable: radioactive

! Isotopes: **same protons, different neutrons**. Some isotopes have an **unstable** nucleus: it gives out radiation. You will meet this in Lesson 5.

Hydrogen's three isotopes



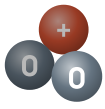
1 p, 0 n

hydrogen-1



1 p, 1 n

hydrogen-2 (deuterium)

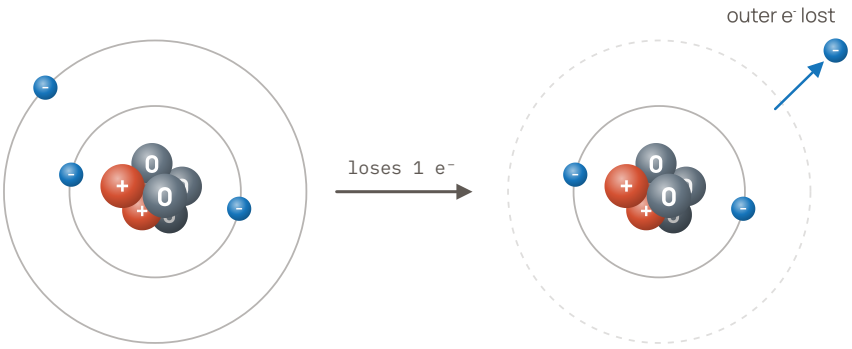


1 p, 2 n

hydrogen-3 (tritium): unstable

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Ions



lithium atom
3 p, 3 e → charge 0

lithium ion, Li^+
3 p, 2 e → charge +1

The outer electrons are furthest from the nucleus and are easiest to remove. If an atom **loses one or more outer electrons**, it has more protons than electrons, so it becomes a **positive ion**.

The nucleus does not change, so it is still the same element. (In chemistry you also meet negative ions, which form when atoms gain electrons.)

WORKED EXAMPLE

A magnesium atom, $^{25}_{12}Mg$, loses 2 electrons. Find the numbers of protons, neutrons and electrons, and the charge on the ion.

PROTONS	12 (unchanged)
NEUTRONS	25 − 12 = 13 (unchanged)
ELECTRONS	12 − 2 = 10
CHARGE	**+12 − 10 = +2, so Mg^{2+}**

YOUR TURN

An aluminium atom, $^{27}_{13}Al$, loses 3 electrons. Find the numbers of each particle and the charge.

PROTONS	13
NEUTRONS	14
ELECTRONS	13 − 3 = 10
CHARGE	+3, so Al^{3+}

PRACTISE

Try it

1 Are these two atoms isotopes of the same element? Circle **Yes** or **No** for each pair.

A $^{12}_{6}C$ $^{14}_{6}C$ — (Yes) No

B $^{14}_{6}C$ $^{14}_{7}N$ — Yes (No)

C $^{35}_{17}Cl$ $^{37}_{17}Cl$ — (Yes) No

D $^{40}_{18}Ar$ $^{40}_{20}Ca$ — Yes (No)

Explain your answer for pair B.

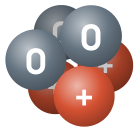
Different numbers of protons (6 and 7), so they are different elements, even though the mass numbers are the same.

2 SUPPORTED Chlorine has two common isotopes. Complete the table.

Isotope	Symbol	Protons	Neutrons	Electrons
chlorine-35	$^{35}_{17}Cl$	17	18	17
chlorine-37	$^{37}_{17}Cl$	17	20	17

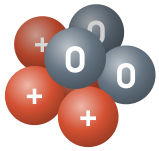
3 The diagrams show three nuclei. Which two are isotopes of the same element? Explain how you know.

P



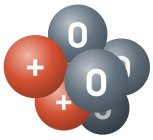
3 p, 3 n

Q



4 p, 3 n

R



3 p, 4 n

P and R: both have 3 protons (the same element, lithium) but different numbers of neutrons (3 and 4).

4 Complete the sentences.

When an atom loses an outer electron it becomes a ___positive___ ion. The number of ___protons___ does not change, so it is still the same ___element___. The ion has more ___protons___ than ___electrons___.

WORD BANK: positive | negative | protons | neutrons | electrons | element

5 ON YOUR OWN Complete the table. Write *atom* or *ion*, and the overall charge.

Protons	Neutrons	Electrons	Atom or ion?	Overall charge
11	12	11	atom	0
11	12	10	ion	+1
20	20	18	ion	+2
8	8	8	atom	0

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6 Explain what is wrong with Ava's statement.

Ava: A lithium atom becomes a Li^+ ion by losing a proton, so its charge goes up by one.

The atom loses an electron, not a proton; the nucleus does not change. Losing a negative electron leaves one more proton than electrons, so the charge is +1. If it lost a proton it would become a different element (helium).

7 CHALLENGE Uranium-235, $^{235}_{92}U$, and uranium-238, $^{238}_{92}U$, are both found in uranium ore. Explain why they are isotopes. Give one way their atoms are the same and one way they are different.

Same number of protons (92), so the same element, but different numbers of neutrons (143 and 146). Same: 92 protons / 92 electrons / same chemistry. Different: number of neutrons / mass number / how stable the nucleus is.

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: 'Similarities and differences' needs both: give at least one of each, with numbers.

8 Carbon has two isotopes, $^{12}_{6}C$ and $^{14}_{6}C$. Describe the similarities and differences between the nuclei of these two isotopes. **[3 marks]**

Both have 6 protons (1). Carbon-12 has 6 neutrons and carbon-14 has 8 neutrons (1). So they have different mass numbers / carbon-14 is unstable (1).

9 Explain how a sodium atom can become a sodium ion, Na^+. **[2 marks]**

It loses one outer electron (1). It now has 11 protons and 10 electrons, so the overall charge is +1 (1).

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Isotopes: same number of protons, different numbers of neutrons.
- Isotopes of an element have the same atomic number but different mass numbers.
- Some isotopes are unstable (radioactive), e.g. carbon-14.
- An atom that loses outer electrons becomes a positive ion.
- Forming an ion does not change the nucleus, so the element stays the same.

COVER THE PAGE AND ANSWER

1 What is the same in two isotopes of an element?

2 What is different in two isotopes of an element?

3 How does an atom become a positive ion?

4 An atom with 12 protons loses 2 electrons. What is its charge?

I CAN…

- explain what isotopes are
- use nuclear notation to compare isotopes
- describe how a positive ion forms
- work out the charge on an ion

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REFERENCE

Toolkit.

The facts and methods you have met so far. Learn these: exam questions expect you to recall them without a data sheet.

The three particles

Lesson 1

Particle	Charge	Mass	Where
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	very small	energy levels

Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ 1×10^{-10} m; nucleus radius < 1/10 000 of that

Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

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REFERENCE

Key words from this booklet.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

Atom: The smallest part of an element. It has a tiny nucleus with electrons around it and no overall charge.

Atomic number: The number of protons in the nucleus. It decides the element.

Electron: A tiny particle with a relative charge of −1, found in energy levels around the nucleus.

Ion: An atom that has lost (or gained) electrons, so it has an overall charge.

Isotopes: Atoms of the same element with different numbers of neutrons.

Mass number: The total number of protons and neutrons in the nucleus.

Neutron: A particle in the nucleus with no charge and a relative mass of 1.

Nucleus: The tiny centre of an atom, containing protons and neutrons and almost all the mass.

Proton: A particle in the nucleus with a relative charge of +1 and a relative mass of 1.

CHECK YOURSELF

Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

LESSON 1: INSIDE THE ATOM

DO NOW

1. Atoms (tiny particles).
2. Protons, neutrons and electrons.
3. A substance made of only one type of atom.

COVER THE PAGE

1. 0
2. 1×10^{-10} m
3. The number of protons + neutrons.
4. 31 − 15 = 16

LESSON 2: ISOTOPES AND IONS

DO NOW

1. 0
2. 31 − 15 = 16
3. It has the same number of electrons as protons.

COVER THE PAGE

1. The number of protons (atomic number).
2. The number of neutrons (mass number).
3. It loses one or more outer electrons.
4. +2

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